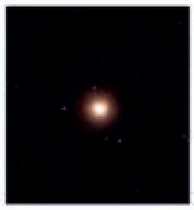
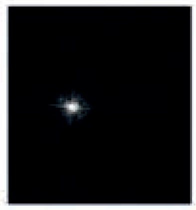
VIEW FROM EARTH

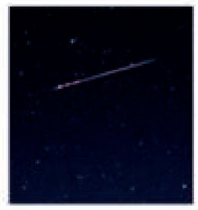
It is impossible to tell how large a star or planet is by looking at it from Earth, because some are huge but very far away. The Sun's diameter is 400 times that of the Moon, but it is also about 400 times further away.

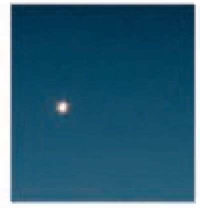

SUN


MOON


MARS


POLARIS

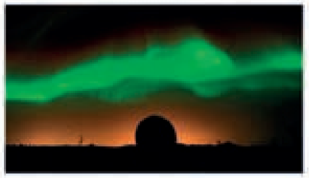

METEOR


VENUS


SATURN

LIGHTS IN THE SKY

Sometimes we can see the interaction of light and magnetism in the skies through colourful light displays such as the northern lights.


NORTHERN LIGHTS
Also known as the aurora borealis, this light display is caused by particles from the Sun hitting Earth's magnetic field.


SOUTHERN LIGHTS
Also known as the aurora australis, this is similar to the northern lights but takes place above Earth's southern hemisphere.


MOONDOG
A moondog appears as a halo around the Moon. It is caused by the refraction of moonlight on ice crystals in clouds.


SUNDOG
Patches of sunlight appear at either side of the Sun. They are caused by sunlight refracting off ice crystals in clouds.

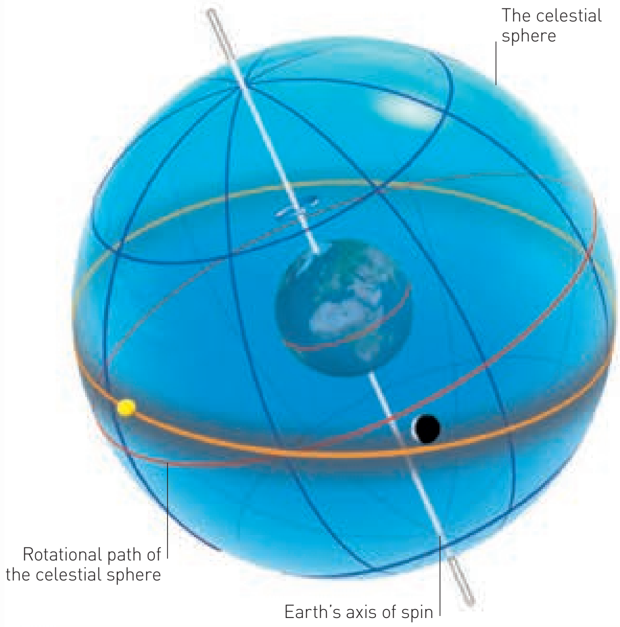
LIFE OUT THERE

The SETI (search for extraterrestrial intelligence) project was set up in 1960 to search for signs of life beyond Earth. Its powerful radio telescopes scan the skies but have not picked up an artificial (non-natural) radio signal so far.


SETI TELESCOPES

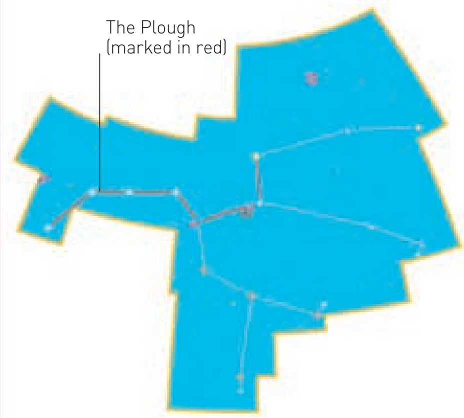
THE CELESTIAL SPHERE

The celestial sphere is an imaginary sphere around Earth. Any sky object can be mapped on to this sphere. Because Earth rotates, the celestial sphere appears to rotate. Like Earth, it has north and south poles and is divided into hemispheres by an equator.



CONSTELLATIONS

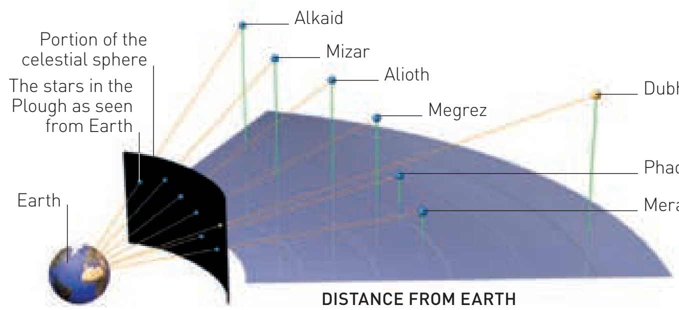
Stargazers in ancient times named groups of stars after mythical beings and animals. These star patterns are called constellations and we still use them today to find the stars. There are 88 constellations in total, and each one is only visible at certain times and from certain places.



URSA MAJOR
This constellation is also known as the Great Bear. It contains an asterism (smaller group of stars) known as the Plough, or the Big Dipper.

LINE OF SIGHT

Wherever you stand on Earth, you can see a portion of the celestial sphere. For example, the Plough seems to take a fixed shape, but it is actually formed by stars moving far out in space, all at different distances from the Earth.



DISTANCE FROM EARTH


Uranus

1781
German-born astronomer, William Herschel discovers Uranus, a planet beyond Saturn, doubling the size of the known Solar System.

1933
American physicist Karl Jansky records the first radio-wave signals from space, which he concludes are from the Milky Way.

1992
Astronomers discover the first extra-solar planets (exoplanets).


Infrared Astronomical Satellite

2006
The International Astronomical Union defines the properties of a "planet" and in doing so demotes Pluto from a planet to a dwarf planet.

2018

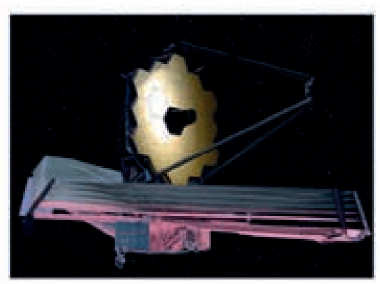

Ceres

1801
Italian astronomer Giuseppe Piazzi comes across a rocky body orbiting between Mars and Jupiter. Named Ceres, this is the largest object in the asteroid belt, and is classified as a dwarf planet.

1843
German amateur astronomer Samuel Heinrich Schwabe observes that sunspots (areas of lower temperature) follow regular cycles.


Sunspots

1922
American astronomer Edwin Hubble works out that there are more galaxies in the Universe than the Milky Way, and that they are moving apart – the Universe is expanding.


James Webb Space Telescope

2018
The James Webb Space Telescope (JWST) is a space observatory scheduled to launch in October 2018. It is a successor to the Hubble Space Telescope and will offer the clearest images ever seen of objects in space.